

Simple linear regression

- The sample: $\{(x_i, y_i)\}_{i=1}^n$

independent
variable

dependent
variable

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2)$$

random deviation or random error

Multiple linear regression with k independent variables. $k > 1$

- The sample: $\{(x_{1i}, x_{2i}, \dots, x_{ki}, y_i)\}_{i=1}^n$



A red arrow points from the x terms in the sample notation to the 'independent variables' box. A blue arrow points from the y term to the 'dependent variable' box.

independent
variables

dependent
variable

$$Y = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k}_{\text{True regression line}} + \varepsilon \quad \varepsilon \sim N(0, \sigma^2)$$

True regression line

random deviation or random error

Summary of Multiple LR

The sample:

$$\{(x_{1i}, x_{2i}, \dots, x_{ki}, y_i)\}_{i=1}^n$$

$$Y = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k}_{\text{True regression line}} + \varepsilon \quad \varepsilon \sim N(0, \sigma^2)$$

- Significance of LR model
 - $H_0: \beta_i = 0$ for all i
 - $H_a: \beta_i \neq 0$ for at least one i
 - F-test for the whole model
 - t-tests for individual independent variables. $H_0: \beta_i = 0$
 $H_a: \beta_i \neq 0$
- Percentage of variability explained by the LR model
 - R^2 : no reference to df
 - Adjusted R^2 : adjusted according to df so as to minimize MSE

Confounding and Collinearity

- *Fitting separate models to each variable is not the same as fitting the multivariate model.*
- An example is an ideal way to see that confounding and collinearity are something that we need always consider and that justify the use of multiple regression models.

Example

A total of 255 gas wells underwent hydraulic fracturing to increase production. In this treatment, a mixture of fluid and sand is pumped into the well. The sand keeps the cracks in the rock open, thereby increasing the flow of gas.

The main questions are these:

- Does increasing the volume of fluid pumped increase the production of the well?
- Does increasing the volume of sand increase the production of the well?

Example

Other things being equal, deeper wells produce more gas, because they provide more surface through which the gas can permeate.

For this reason, it is appropriate to express all variables in units per foot of depth of the well. Thus production is measured in units of ft^3 of gas per ft of depth, fluid is measured in units of gal/ft, and sand is measured in units of lb/ft.

Both fluid and sand appear to be strongly related to production.

Simple Linear Regression Models: Fluid & Sand

In Prod = -0.444 + 0.798 In Fluid

Predictor	Coef	SE Coef	T	P
Constant	-0.4442	0.5853	-0.76	0.449
In Fluid	0.79833	0.08010	9.97	0.000

S = 0.7459

R - Sq = 28.2%

R - Sq(adj) = 27.9%

In Prod = -0.778 + 0.748 In Sand

Predictor	Coef	SE Coef	T	P
Constant	-0.7784	0.6912	-1.13	0.261
In Sand	0.74751	0.08381	8.92	0.000

S = 0.7678

R - Sq = 23.9%

R - Sq(adj) = 23.6%

Multiple Regression

$$\text{In Prod} = -0.729 + 0.670 \text{ In Fluid} + 0.148 \text{ In Sand}$$

Predictor	Coef	SE Coef	T	P
Constant	-0.7288	0.6791	-1.08	0.279
In Fluid	0.6701	0.1687	3.97	0.000
In Sand	0.1481	0.1714	0.86	0.389

S = 0.7463

R - Sq = 28.4%

R - Sq(adj) = 27.8%

What do you think is happening here?

Confounding: If we increase the volume of fluid, then we also increase the volume of sand.

Understanding confounding and conclusion

- Multiple regression model:
 - R^2 and $\text{Adj } R^2$ are not similar to those in the simple regression models.
 - Sand is not significant any more.
- This indicates that there are other important factors affecting production that have not been included in the models.

Collinearity

- When two independent variables are very strongly correlated, multiple regression may not be able to determine which is the important one.
- In this case, the variables are said to be **collinear**. The word *collinear* means to lie on the same line, and when two variables are highly correlated, their scatterplot is approximately a straight line.
- The word *multicollinear* is sometimes used as well, meaning that multiple variables are highly correlated with each other.
- When collinearity is present, the set of independent variables is sometimes said to be **ill-conditioned**.

Comments

- Sometimes two variables are so correlated that multiple regression cannot determine which is responsible for the linear relationship with y .
- In general, there is not much that can be done when variables are collinear.
- The only way to fix the situation is to collect more data, including some values for the independent variables that are not on a straight line.

Example

Simple Linear Regression

x_1	x_2	y
0.1	0.3	3.6
0.2	0.2	0.3
0.6	1.4	6.0
1.4	3.4	10.6
2.0	5.2	8.4
2.0	5.5	11.8
2.1	5.5	12.7
2.1	5.3	6.8
2.8	7.4	9.9
3.6	9.4	16.7
4.2	10.3	16.3
4.5	11.4	19.9
4.7	11.3	20.2
5.3	13.6	22.9
6.1	15.3	26.6
6.8	17.4	28.1
7.5	18.5	31.0
8.2	20.4	28.8
8.5	21.3	32.4
9.4	23.3	35.0

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.8988	0.8224	3.525	0.00242
X1	3.5326	0.1652	21.384	3.03e-14

Residual standard error: 2.08 on 18 degrees of freedom

Multiple R-squared: 0.9621, Adjusted R-squared: 0.96

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.74315	0.80900	3.391	0.00326
X2	1.42024	0.06485	21.902	2e-14

Residual standard error: 2.033 on 18 degrees of freedom

Multiple R-squared: 0.9638, Adjusted R-squared: 0.9618

ExamExample

Multiple Linear Regression

x_1	x_2	y
0.1	0.3	3.6
0.2	0.2	0.3
0.6	1.4	6.0
1.4	3.4	10.6
2.0	5.2	8.4
2.0	5.5	11.8
2.1	5.5	12.7
2.1	5.3	6.8
2.8	7.4	9.9
3.6	9.4	16.7
4.2	10.3	16.3
4.5	11.4	19.9
4.7	11.3	20.2
5.3	13.6	22.9
6.1	15.3	26.6
6.8	17.4	28.1
7.5	18.5	31.0
8.2	20.4	28.8
8.5	21.3	32.4
9.4	23.3	35.0

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.7248	0.8488	3.210	0.00513
x1	-0.4895	4.4596	-0.110	0.91388
x2	1.6167	1.7914	0.903	0.37939

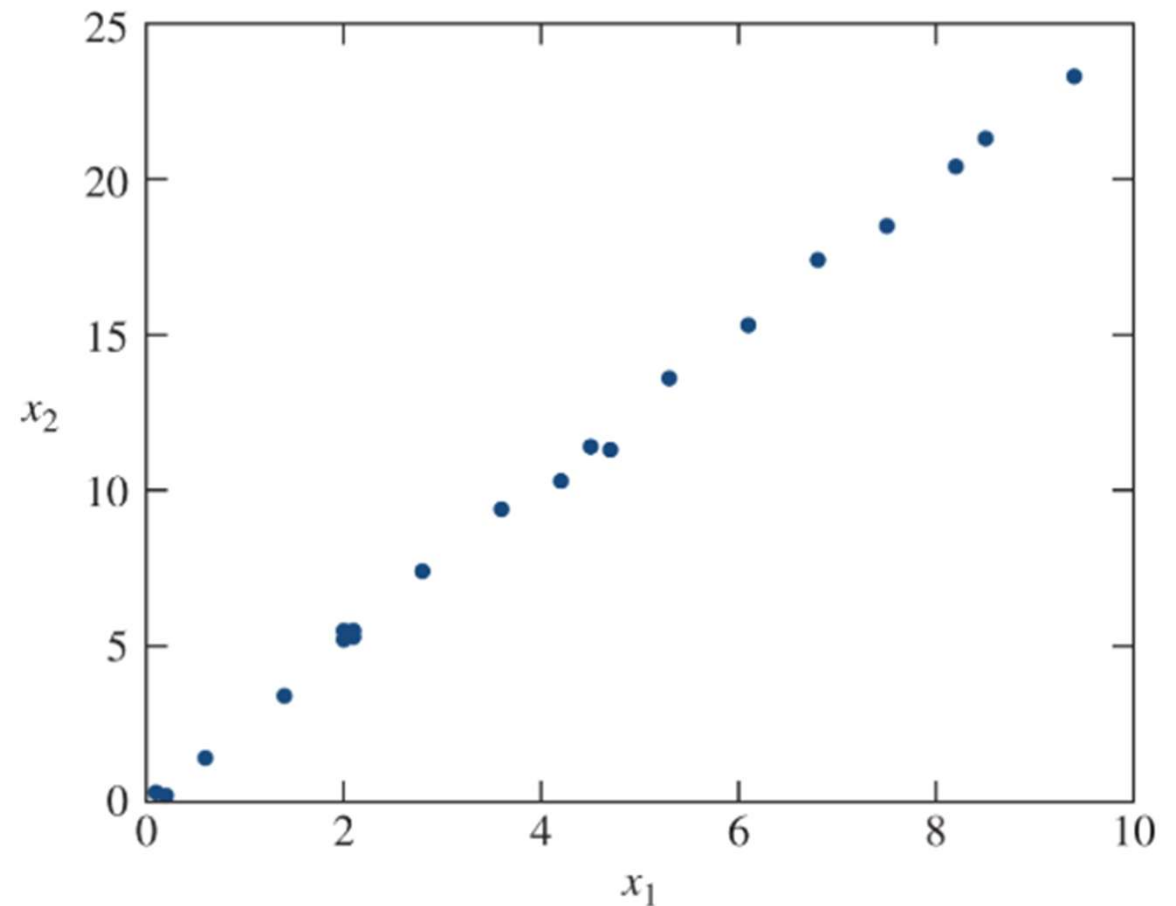
Residual standard error: 2.091 on 17 degrees of freedom

Multiple R-squared: 0.9639, Adjusted R-squared: 0.9596

Understanding the Example

Correlation between x_1 & x_2

x_1	x_2	y
0.1	0.3	3.6
0.2	0.2	0.3
0.6	1.4	6.0
1.4	3.4	10.6
2.0	5.2	8.4
2.0	5.5	11.8
2.1	5.5	12.7
2.1	5.3	6.8
2.8	7.4	9.9
3.6	9.4	16.7
4.2	10.3	16.3
4.5	11.4	19.9
4.7	11.3	20.2
5.3	13.6	22.9
6.1	15.3	26.6
6.8	17.4	28.1
7.5	18.5	31.0
8.2	20.4	28.8
8.5	21.3	32.4
9.4	23.3	35.0



Model Selection

- There are many situations in which a large number of independent variables have been measured, and we need to decide which of them to include in a model.
- This is the problem of **model selection**, and it is a difficult one.
- In practice, model selection often proceeds by ad hoc methods, guided by whatever physical intuition may be available.
- We will not attempt a complete discussion of this extensive and difficult topic. Instead, we will be content to state some basic principles and to present some examples.

Model Selection and Occam's Razor

- Good model selection rests on this basic principle known as Occam's razor:
“The best scientific model is the **simplest** model that explains the observed data.”
- In terms of linear models, Occam's razor implies the **Principle of Parsimony**:
“A model should contain the smallest number of variables necessary to fit the data.”

Some Exceptions

1. A linear model should always contain an intercept, unless physical theory dictates otherwise.
2. If a power X^n of a variable is included in the model, all lower powers $X, X^2, \dots, X^{n-1}$ should be included as well, unless physical theory dictates otherwise.
3. If a product $x_i x_j$ of two variables is included in a model, then the variables x_i and x_j should be included separately as well, unless physical theory dictates otherwise.

Notes

- Models that include only the variables needed to fit the data are called **parsimonious** models. Much of the practical work of multiple regression involves the development of parsimonious models.
- Adding a new variable to a model can substantially change the coefficients of the variables already in the model.

EXAMPLE

TABLE 8.3 Feed rates and power for industrial jaw crushers

Feed Rate (100 tons/h)	Power (kW)
0.10	11
1.55	60
3.00	40
3.64	150
0.38	69
1.59	77
4.73	83
0.20	15
2.91	84
0.36	30
0.14	16
0.91	30
4.27	150
4.36	144
0.91	45
0.59	12
0.27	24
0.55	49
0.68	45
4.27	150
3.64	100
1.36	58
2.36	45
2.95	75
1.09	44
0.91	58
2.91	149

$$\text{Power} = \beta_0 + \beta_1 \text{FeedRate} + \varepsilon$$

Coefficients:				
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	21.028	8.038	2.616	0.0149
FeedRate	24.595	3.338	7.368	1.02e-07

$$\text{Power} = \beta_0 + \beta_1 \text{FeedRate} + \beta_2 \text{FeedRate}^2 + \varepsilon$$

Coefficients:				
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	19.3441	11.5571	1.674	0.1072
FeedRate	27.4678	14.3093	1.920	0.0669
FeedRate^2	-0.6387	3.0899	0.207	0.8380

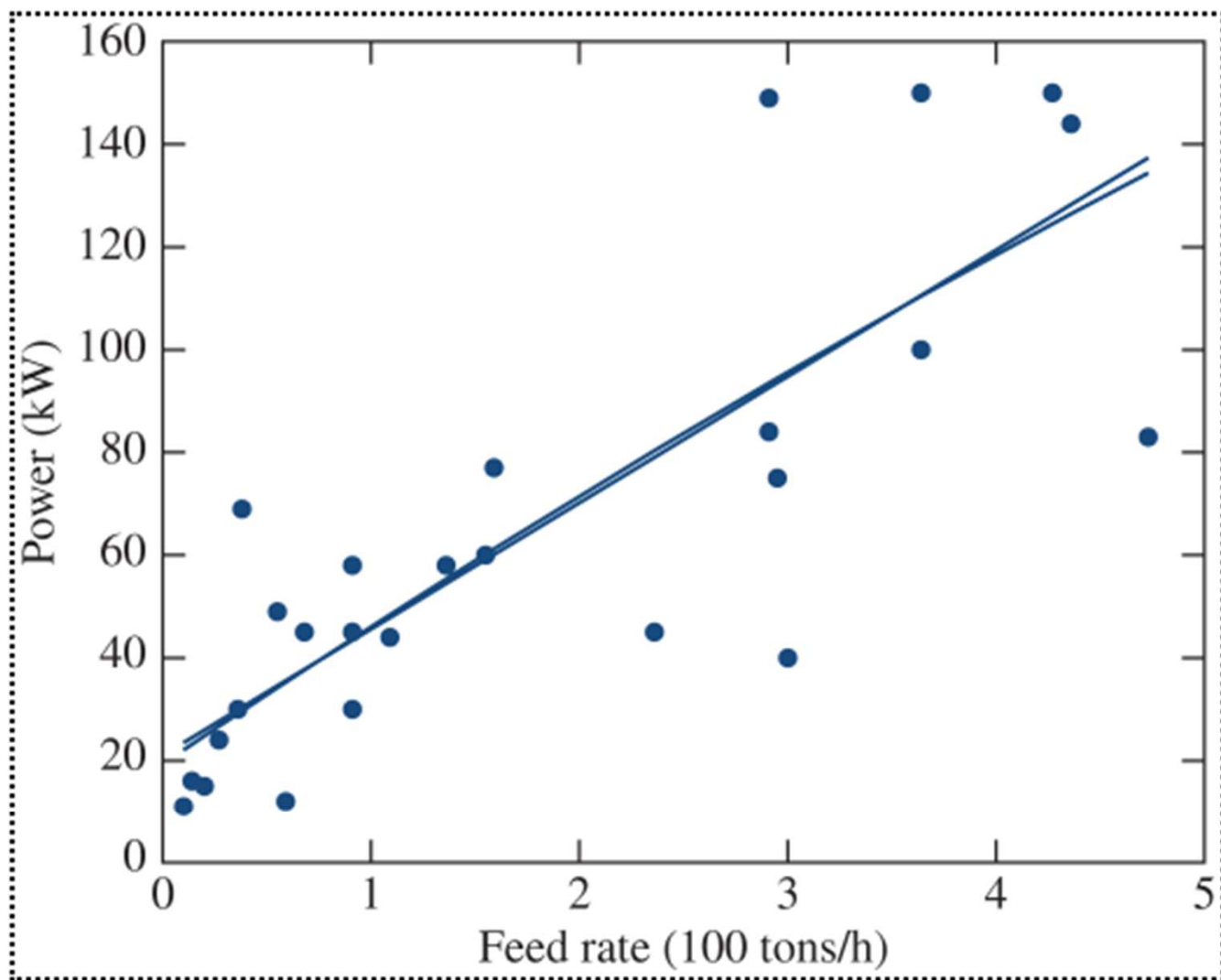


FIGURE 8.7 Scatterplot of power versus feed rate for 27 industrial jaw crushers. The least-squares line and best-fitting quadratic model are both superimposed. The two curves are practically identical, which reflects the fact that the coefficient of FeedRate^2 in the quadratic model does not differ significantly from 0.

Can a Variable Be Dropped?

- It often happens that one has formed a model that contains a large number of independent variables, and one wishes to determine whether a given subset of them may be dropped from the model without significantly reducing the accuracy of the model.
- Assume that we know that the model

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \beta_{k+1} x_{k+1i} + \cdots + \beta_p x_{pi} + \varepsilon_i$$

is correct. We will call this the **“full” model**.

Null Hypothesis

- We wish to test the null hypothesis

$$H_0: \beta_{k+1} = \cdots = \beta_p = 0$$

- If H_0 is true, the model will remain correct if we drop the variables $X_{k+1}, \dots, X_p$, so we can replace the full model with the following **reduced model**:

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \varepsilon_i$$

$$H_0: \beta_{k+1} = \cdots = \beta_p = 0$$

Test Statistic

- To develop a test statistic for H_0 , we begin by computing the error sums of squares for both the full and reduced models.
- We call them SS_{full} and SS_{reduced} , respectively.
- The number of degrees of freedom for SS_{full} is $n - p - 1$, and for SS_{reduced} is $n - k - 1$.

$$H_0: \beta_{k+1} = \cdots = \beta_p = 0$$

Test Statistic

- The test statistic is

$$f = \frac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}})/(p - k)}{\text{SSE}_{\text{full}}/(n - p - 1)} \sim F_{p-k, n-(p+1)}$$

- If H_0 is true, both numerator and denominator are estimates of σ^2 , so f tends to be close to 1.

If H_0 is false, the quantity $\text{SSE}_{\text{reduced}}$ tends to be larger, so f tends to be larger.

Comments

- The method we have just described is very useful in practice for developing parsimonious models by removing unnecessary variables.
- However, the conditions under which it is formally valid are seldom met in practice.

Why Conditions May Not Be Met

1. It is rarely the case that the full model is correct; there will be nonrandom quantities that affect the value of the dependent variable y that are not accounted for by the independent variables.
2. For the method to be formally valid, the subset of variables to be dropped must be determined independently of the data. This is usually not the case.

Comments

- More often, a large model is fit, some of the variables are seen to have fairly large P -values, and the F test is used to decide whether to drop them from the model.
- This is a useful technique in practice, but, like most methods of model selection, it should be seen as an informal tool rather than a rigorous theory-based procedure.
- As such, it is possible to find a different final model that might have been just as good as the one we actually found.
- In practice, there are often many models that fit the data about equally well; there is no single “correct” model.

Example

Mobile Ad Hoc Computer Networks

- Dependent variable:
Data overhead (# bytes of information to be transmitted along the message to get them to the right places)
- Independent variables:
Speed, Pause time, Link change rate

TABLE Data overhead, speed, pause time, and link change rate for a mobile computer network

Speed (m/s)	Pause Time (s)	LCR (100/s)	Data Overhead (kB)
5	10	9.426	428.90
5	20	8.318	443.68
5	30	7.366	452.38
5	40	6.744	461.24
5	50	6.059	475.07
10	10	16.456	446.06
10	20	13.281	465.89
10	30	11.155	477.07
10	40	9.506	488.73
10	50	8.310	498.77
20	10	26.314	452.24
20	20	19.013	475.97
20	30	14.725	499.67
20	40	12.117	501.48
20	50	10.284	519.20
30	10	33.009	445.45
30	20	22.125	489.02
30	30	16.695	506.23
30	40	13.257	516.27
30	50	11.107	508.18
40	10	37.823	444.41
40	20	24.140	490.58
40	30	17.700	511.35
40	40	14.064	523.12
40	50	11.691	523.36

F Table

		v_1		
v_2	α	1	2	3
10	0.100	3.29	2.92	2.73
	0.050	4.96	4.10	3.71
	0.010	10.04	7.56	6.55
	0.001	21.04	14.91	12.55
11	0.100	3.23	2.86	2.66
	0.050	4.84	3.98	3.59
	0.010	9.65	7.21	6.22
	0.001	19.69	13.81	11.56
12	0.100	3.18	2.81	2.61
	0.050	4.75	3.89	3.49
	0.010	9.33	6.93	5.95
	0.001	18.64	12.97	10.80
13	0.100	3.14	2.76	2.56
	0.050	4.67	3.81	3.41
	0.010	9.07	6.70	5.74
	0.001	17.82	12.31	10.21
14	0.100	3.10	2.73	2.52
	0.050	4.60	3.74	3.34
	0.010	8.86	6.51	5.56
	0.001	17.14	11.78	9.73
15	0.100	3.07	2.70	2.49
	0.050	4.54	3.68	3.29
	0.010	8.68	6.36	5.42
	0.001	16.59	11.34	9.34
16	0.100	3.05	2.67	2.46
	0.050	4.49	3.63	3.24
	0.010	8.53	6.23	5.29
	0.001	16.12	10.97	9.01
17	0.100	3.03	2.64	2.44
	0.050	4.45	3.59	3.20
	0.010	8.40	6.11	5.18
	0.001	15.72	10.66	8.73

Full model

The regression equation is

$$\text{Overhead} = 368 + 3.48 \text{ LCR} + 3.04 \text{ Speed} + 2.29 \text{ Pause} - 0.0122 \text{ Speed*Pause} \\ - 0.1041 \text{ LCR}^2 - 0.0313 \text{ Speed}^2 - 0.0132 \text{ Pause}^2$$

Predictor	Coef	SE Coef	T	P
Constant	367.96	19.40	18.96	0.000
LCR	3.477	2.129	1.63	0.121
Speed	3.044	1.591	1.91	0.073
Pause	2.2924	0.6984	3.28	0.004
Speed*Pa	-0.01222	0.01534	-0.80	0.437
LCR^2	-0.10412	0.03192	-3.26	0.005
Speed^2	-0.03131	0.01906	-1.64	0.119
Pause^2	-0.01318	0.01045	-1.26	0.224

S = 5.72344

R-Sq = 97.2%

R-Sq(adj) = 96.1%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	7	19567.5	2795.4	85.33	0.000
Residual Error	17	556.9	32.8		
Total	24	20124.3			

Reduced model - 1

ν_2	α	1	2	3
17	0.100	3.03	2.64	2.44
	0.050	4.45	3.59	3.20
	0.010	8.40	6.11	5.18
	0.001	15.72	10.66	8.73

The regression equation is

$$\text{Overhead} = 359 + 6.69 \text{ LCR} + 0.777 \text{ Speed} + 1.67 \text{ Pause} - 0.156 \text{ LCR}^2$$

Predictor	Coef	SE Coef	T	P
Constant	359.22	13.01	27.61	0.000
LCR	6.695	1.156	5.79	0.000
Speed	0.7766	0.2054	3.78	0.001
Pause	1.6729	0.1826	9.16	0.000
LCR^2	-1.5572	0.02144	-7.26	0.000

S = 6.44304

R-Sq = 95.9%

R-Sq(adj) = 95.0%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	4	19294.1	4823.5	116.19	0.000
Residual Error	20	830.3	41.5		
Total	24	20124.3			

$$\text{Test statistic: } \frac{(SSE_{\text{reduced}} - SSE_{\text{full}}) / (p - k)}{\frac{SSE_{\text{full}}}{n - p - 1}} = \frac{(830.3 - 556.9) / (7 - 4)}{\frac{556.9}{17}} = 2.78$$

$$\sim F_{p-k, n-p-1} = F_{3, 17}$$

Reduced model - 2

The regression equation is

$$\text{Overhead} = 373 + 4.80 \text{ LCR} + 1.99 \text{ Speed} + 1.45 \text{ Pause} - 0.0212 \text{ Speed}^2$$

ν_2	α	1	2	3
17	0.100	3.03	2.64	2.44
	0.050	4.45	3.59	3.20
	0.010	8.40	6.11	5.18
	0.001	15.72	10.66	8.73

Predictor	Coef	SE Coef	T	P
Constant	372.60	16.93	22.00	0.000
LCR	4.799	1.935	2.48	0.023
Speed	1.993	1.023	1.95	0.066
Pause	1.4479	0.2587	5.60	0.000
LCR^2	-0.12345	0.03400	-3.63	0.002
Speed^2	-0.02120	0.01746	-1.21	0.240

S = 6.36809

R-Sq = 96.2%

R-Sq(adj) = 95.2%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	5	19353.8	3870.8	95.45	0.000
Residual Error	19	770.5	40.6		
Total	24	20124.3			

$$\text{Test statistic: } \frac{(SSE_{\text{reduced}} - SSE_{\text{full}}) / (p - k)}{\frac{SSE_{\text{full}}}{n - p - 1}} = \frac{(770.5 - 556.9) / (7 - 5)}{\frac{556.9}{17}} = 3.26$$

$$\sim F_{p-k, n-p-1} = F_{2, 17}$$

Reduced model - 3

The regression equation is

$$\text{Overhead} = 345 + 6.484 \text{ LCR} + 0.707 \text{ Speed} + 2.85 \text{ Pause} - 0.1 \text{ LCR}^2 - 0.0183 \text{ Pause}^2$$

ν_2	α	1	2	3
17	0.100	3.03	2.64	2.44
	0.050	4.45	3.59	3.20
	0.010	8.40	6.11	5.18
	0.001	15.72	10.66	8.73

Predictor	Coef	SE Coef	T	P
Constant	345.42	13.19	26.20	0.000
LCR	6.484	1.050	6.17	0.000
Speed	0.7072	0.1883	3.76	0.001
Pause	2.8537	0.5337	5.35	0.000
LCR^2	-0.14482	0.01996	-7.25	0.000
Pause^2	-0.018334	0.007879	-2.33	0.031

S = 5.83154

R-Sq = 96.8%

R-Sq(adj) = 95.9%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	5	19478.2	3895.6	114.55	0.000
Residual Error	19	646.1	34.0		
Total	24	20124.3			

Test statistic:
$$\frac{(SSE_{reduced} - SSE_{full}) / (p - k)}{\frac{SSE_{full}}{n - p - 1}} = \frac{(646.1 - 556.9) / (7 - 5)}{\frac{556.9}{17}} = 1.36$$

$\sim F_{p-k, n-p-1} = F_{2, 17}$

Questions

Which model is the best?

- a) Check whether the reduced models are plausible, compared to the full model.
- b) Which of the reduced models is the best? Why? Explain.

Hypothesis test: $H_0: \beta_{k+1} = \dots = \beta_p = 0$

- Test Statistic: f

$$\frac{(SSE_{reduced} - SSE_{full}) / (p - k)}{SSE_{full} / (n - p - 1)} \sim F_{p-k, n-p-1}$$

- If H_0 is true, both numerator and denominator are estimates of σ^2 , so f tends to be close to 1.
- If H_0 is false, the quantity $SSE_{reduced}$ tends to be larger, so f tends to be larger.

Summary

$$\text{Full model: } F = \frac{(SST - SSE)/p}{SSE/(n-p-1)} \sim F_{p, n-p-1}$$

Model	Null Hypotheses	# Par	F-Statistic	P-value	Test result	Conclusion
Full	All $\beta_i = 0$ $i \geq 1$	7	85.33	0.000	Reject H0 at all plausible α values	Full model: plausible
Red-1	$\beta_4 = \beta_6 = \beta_7 = 0$	4	$\frac{(830.3 - 556.9)/(7 - 4)}{556.9/17} = 2.78$	0.05 < P < 0.10	Cannot reject H0 at $\alpha=0.05$ Reject H0 at $\alpha=0.10$	Red-1: Barely plausible
Red-2	$\beta_4 = \beta_7 = 0$	5	$\frac{(770.5 - 556.9)/(7 - 5)}{556.9/17} = 3.26$	0.05 < P < 0.10	Cannot reject H0 at $\alpha=0.05$ Reject H0 at $\alpha=0.10$	Red-2: Barely plausible
Red-3	$\beta_4 = \beta_6 = 0$	5	$\frac{(646.1 - 556.9)/(7 - 5)}{556.9/17} = 1.36$	P > 0.10	Cannot reject H0 at all plausible α values	Red-3: plausible

$$\text{Reduced model: } f = \frac{(SSE_{\text{reduced}} - SSE_{\text{full}})/(p-k)}{SSE_{\text{full}}/(n-p-1)} \sim F_{p-k, n-p-1}$$

Methods to find the “best” regression model

- All possible regressions
- Stepwise regression (partial F-tests)
 - Forward selection
 - Backward elimination
- Performed by statistical packages.
- There is not one specific rule to choose the “best” model. You should use your judgement, given all the criteria...

Best Subsets Regression

- Assume that there are p independent variables, $x_1, x_2, \dots, x_p$, that are available to be put in the model.
- Let's assume that we wish to find a good model that contains exactly four independent variables.
- We can simply fit every possible model containing four of the variables, and rank them in order of their goodness-of-fit, as measured by the coefficient of determination, R^2 .
- The subset of four variables that yield the largest value R^2 of is the “best” subset of size four.

Best Subsets Regression

- The process can be repeated for subsets of other sizes, finding the best subsets of size $1, 2, \dots, p$.
- These best subsets can be examined to see which provides a good fit, while being parsimonious.
- The best subsets procedure is computationally intensive.
- When there are a lot of potential independent variables, there are a lot of models to fit. However, for most data sets, we can rely on technology to do our calculations.

Determining “Best”

- Methods have been developed to choose a single model, presumably the “best” of the “best.”
- We describe two of them here, with a caution not to take them too seriously.
- Note that if we simply choose the model with the highest value of R^2 , we will always pick the one that contains all the variables, since the value of R^2 necessarily increases as the number of variables in the model increases.
- The methods for selecting a model involve statistics that adjust the value of R^2 , so as to eliminate this feature.

Determining “Best” – Method 1

- The first is the adjusted R^2 .

Let n denote the number of observations, and let k denote the number of independent variables in the model. The adjusted R^2 :

$$\text{Adjusted } R^2 = R^2 - \left(\frac{k}{n - k - 1} \right) (1 - R^2)$$

Determining “Best” – Method 2

- The adjusted R^2 is always smaller than R^2 , since a positive quantity is subtracted from R^2 .
- As the number of variables k increases, R^2 will increase, but the amount subtracted from it will increase as well.
- The value of k for which the value of adjusted R^2 is a maximum can be used to determine the number of variables in the model, and the best subset of that size can be chosen as the model.

Determining “Best” – Method ₃

- Another commonly used statistic is **Mallows' C_p** .
- To compute this quantity, let n be the number of observations, let p be the total number of independent variables under consideration, and let k be the number of independent variables in a subset.
- As before, let SSE_{full} denote the error sum of squares for the full model containing all p variables, and let SSE_{reduced} denote the error sum of squares for the model containing only the subset of k variables.

Determining “Best” – Method ₄

- Mallows' C_p is defined as:

p : # of all parameters

$$C_p = \frac{(n - p - 1)SSE_{\text{reduced}}}{SSE_{\text{full}}} - (n - 2k - 2)$$

k : # of parameters in a subset

- For models that contain as many independent variables as necessary, the value of C_p is supposed to be approximately equal to the number of variables, including the intercept, in the model.
- To choose a single model, one can either choose the model with the minimum value of C_p , or one can choose the model in which the value of C_p is closest to the number of independent variables in the model.

Class Exercise: Mobile Ad Hoc Computer Networks

- Dependent variable: Data overhead (# bytes of information to be transmitted along the message to get them to the right places)
- Independent variables: Speed, Pause time, Link change rate

Best Subsets Regression
Response is Overhead

Vars	R-Sq	Adj. R-Sq	C-p	s	S p e e d	P a u s e	L C R	S p e e d ^ 2	P a u s e ^ 2	s p e e d * p a u s e	L C R ^ 2
1	73.7	72.5	140.6	15.171						X	
1	54.5	52.6	258.3	19.946		X					
2	82.7	81.2	87.0	12.564		X				X	
2	82.2	80.6	90.3	12.755	X	X					
3	92.9	91.9	26.5	8.2340		X	X				X
3	89.6	88.1	46.9	9.9870	X		X	X			
4	95.9	95.0	10.3	6.4430	X	X	X				X
4	95.4	94.5	13.2	6.7991		X	X	X			X
5	96.8	95.9	6.7	5.8315	X	X	X		X		X
5	96.7	95.8	7.2	5.9074	X	X		X		X	X
6	97.1	96.2	6.6	5.6651	X	X	X	X	X		X
6	97.0	96.0	7.6	5.8164	X	X	X	X		X	X
7	97.2	96.1	8.0	5.7234	X	X	X	X	X	X	X